Poisson regression

The Poisson model used **2,299 complete observations** from places_2025 (1).csv. It removed 844 observations missing one or more selected values. All 11 assigned predictors were included, with mhlth_crude_prev as the response.

1. What are the four significant predictors?

At a significance level of $p < 0.05$, the four significant predictors are shown below. Column names use the snake_case formatting produced by the script.

Predictor	Coefficient	p-value	VIF
depression_crude_prev	0.01365	6.64 x 10^-9	2.22
selfcare_crude_prev	-0.06077	0.00677	30.51
indeplive_crude_prev	0.05216	0.000171	30.95
loneliness_crude_prev	0.01127	0.000201	3.43

2. Which predictor is most significant based on the coefficient?

Following the assignment's coefficient hint, **selfcare_crude_prev** has the largest coefficient in absolute value, **-0.06077**. It has the largest estimated effect per one-percentage-point increase among the significant predictors. Statistically, **depression_crude_prev** has the strongest evidence of an association because it has the smallest p-value. Coefficient size and statistical significance describe different things.

3. Do any significant predictors have a VIF greater than 10?

Yes. **selfcare_crude_prev** has a VIF of **30.51**, and **indeplive_crude_prev** has a VIF of **30.95**. These values indicate substantial multicollinearity, making their individual coefficients harder to interpret reliably.

4. How much variability is captured by the model?

The McFadden pseudo R squared is **0.04356**. This represents approximately a **4.36% improvement in log-likelihood** relative to the intercept-only model. It should not be interpreted as 4.36% of the variance explained because pseudo R squared differs from ordinary linear-regression R squared.

Model limitation: mhlth_crude_prev contains prevalence percentages rather than counts. These results reproduce the requested Poisson demonstration, but the response does not satisfy the count-data requirement for conventional Poisson regression.

Logistic and Poisson Regression Analysis

Logistic regression

The logistic model used 189 observations from birthwt (1).csv. The predictors were smoke, ht, and ui, and the response was low. Predictions used a probability cutoff of 0.50.

Actual category	Predicted 0	Predicted 1	Support
0 - Normal birthweight	119	11	130
1 - Low birthweight	45	14	59

1. What is the model's accuracy?

The accuracy is **70.37%**. The model correctly classified 133 of 189 observations.

2. What is the model's lowest precision?

The lowest precision is **56.00%**, for the low-birthweight category. Of the 25 observations predicted to have low birthweight, 14 actually did.

3. Do all categories have at least 30 support?

Yes. The normal-birthweight category has **130 observations**, and the low-birthweight category has **59 observations**.

4. Based on the pseudo R squared value, is this model good in this scenario?

The McFadden pseudo R squared is **0.0625**, indicating limited improvement over an intercept-only model. The model identifies only **23.73%** of actual low-birthweight observations. Therefore, its usefulness for identifying low birthweight is limited despite its 70.37% overall accuracy.

These classification results use the same observations used to fit the model. They do not measure performance on new data.